

A Literature-Implied Convex-Domain Answer to an FPV Sum Problem

Math discovery packet

June 28, 2026

1 Status

This is a lightweight *literature-implied answer*, not an original solution. The source paper is J. M. Borwein and L. Yao, *Recent progress on Monotone Operator Theory*, arXiv:1210.3401. In Section “Open problems in Monotone Operator Theory”, p. 33 of the arXiv PDF, the paper asks:

Let $A, B : X \rightrightarrows X^*$ be maximally monotone with $\text{dom } A \cap \text{int dom } B \neq \emptyset$. Assume that A is of type (FPV). Is $A + B$ necessarily maximally monotone?

The supporting paper is J. M. Borwein and L. Yao, *Sum theorems for maximally monotone operators of type (FPV)*, arXiv:1305.6691.

2 Identification

The supporting paper proves the following theorem and corollary.

1. Theorem 4.2, the “(FPV) Sum Theorem”, proves maximal monotonicity of $A + B$ when A and B are maximally monotone, $\text{star}(\text{dom } A) \cap \text{int dom } B \neq \emptyset$, and A is of type (FPV).
2. Corollary 4.3, “Convex domain”, states that if $\text{dom } A \cap \text{int dom } B \neq \emptyset$ and A is of type (FPV) with convex domain, then $A + B$ is maximally monotone.

Since convexity of $\text{dom } A$ implies $\text{star}(\text{dom } A) = \text{dom } A$, Corollary 4.3 gives a direct affirmative answer to the source question in the convex-domain subcase.

3 Scope

The original arXiv:1210.3401 question does not assume that $\text{dom } A$ is convex. The supporting paper does not claim to settle that general case: after Corollary 4.3 it explicitly asks whether the star-shaped hypothesis on $\text{dom } A$ in Theorem 4.2 can be relaxed or dropped, and then re-poses the unqualified FPV sum problem.

A later arXiv paper, arXiv:1411.6199, claimed to settle the unqualified question, but the arXiv record states that the paper has been withdrawn by the author because of an error in Lemma 1. This packet therefore records only the valid convex-domain literature subcase.

4 References

- J. M. Borwein and L. Yao, *Recent progress on Monotone Operator Theory*, arXiv:1210.3401.
- J. M. Borwein and L. Yao, *Sum theorems for maximally monotone operators of type (FPV)*, arXiv:1305.6691.
- S. R. Pattanaik and D. K. Pradhan, *Maximality of the sum of the monotone operator of type (FPV) and a maximal monotone operator*, arXiv:1411.6199, withdrawn.